

WEEK 4

SUPERVISED MACHINE LEARNING FOR BUSINESS

Classification • regression • features • labels • probabilities • business decisions

**TEXTBOOK •
CHAPTER 4**

THU • SEP 17, 2026

**GRADUATE • 3
HOURS**



WEEK 4 → WEEK 4

From usable evidence to prediction

Last week: “Is the evidence ready?” Today: “Can labeled examples help us predict a future outcome?”

WEEK 3

Language • evidence •
labels • quality • access



WEEK 4

Historical examples →
model → probability →
business action



WEEK 5

Segmentation •
recommendation •
unsupervised patterns

Supervised learning needs a target outcome: the historical “answer” the model learns from.

CURRENT AI BRIEFING

AI NOW: three signals since our last class

Capability • adoption • governance — what changed, and why should a manager care?

1 CAPABILITY

SEP 10

OpenAI introduced the Agents API, expanding the tooling available for developers to build agentic workflows.

QUESTION: What changes when AI can call tools and execute multi-step work?

2 ADOPTION

SEP 14

Anthropic introduced Claude for Financial Advisors with connectors and workflow skills for research, preparation, and documentation.

QUESTION: Why are vertical, role-specific AI products important?

3 GOVERNANCE

SEP 10

Anthropic published a threat-intelligence report describing malicious attempts to use Claude.

QUESTION: What controls are needed when model capability increases?

Sources: OpenAI Product News (Sep. 10, 2026) • Claude/Anthropic (Sep. 10 & Sep. 14, 2026)

TODAY

Week 4 outcomes + graded work

One lecture block, one applied lab

LEARN

- features vs. label
- classification vs. regression
- training vs. prediction
- probability + threshold
- model errors

At home individual activity

Applied Lab 1 — Customer Churn

Interpret a pre-trained model, apply the 0.60 threshold, identify errors, and make a managerial recommendation.

Homework

12 short-answer questions based on the textbook + Week 4 lecture.

PART I

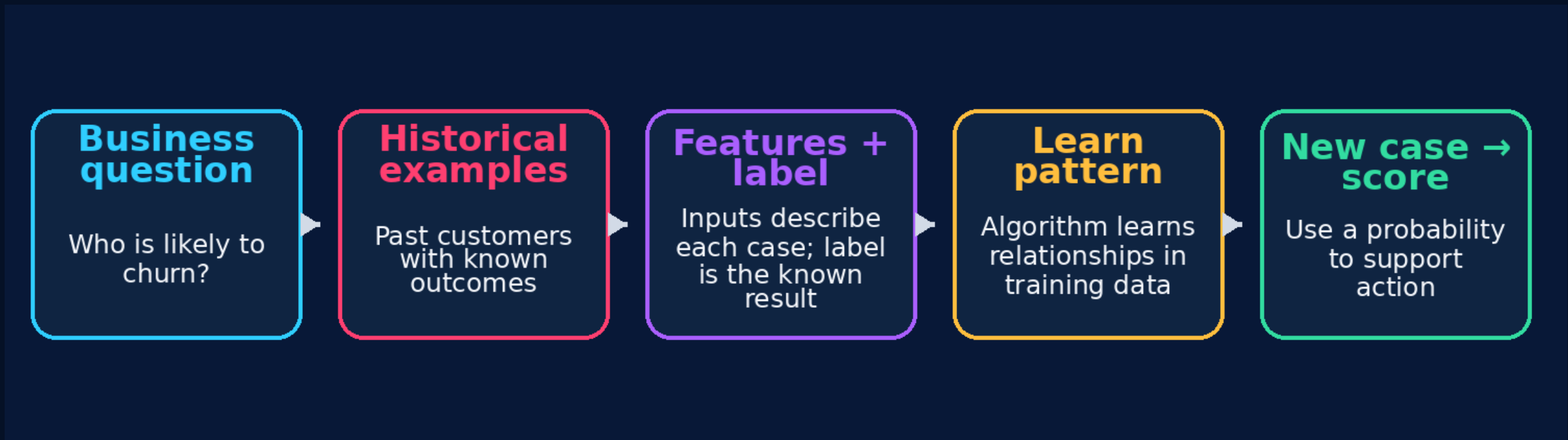
SUPERVISED LEARNING — THE CORE MODEL

Known historical outcomes → learned pattern → new prediction → business decision

ONE MENTAL MODEL

Supervised learning in one picture

Historical examples with known outcomes are used to learn a mapping from inputs to a target.



VOCABULARY

One row = one example

Features describe the case. The label is the historical outcome we want the model to learn.

Customer	Tenure	Monthly charge	Tickets	Satisfaction	Contract	Churn?
C001	2 mo	\$95	5	1 / 5	Month-to-month	Yes

FEATURE

An input used to describe the customer — e.g., tenure or support tickets.

LABEL / TARGET

The known historical outcome we want to predict — here, churned: Yes / No.

NOT A FEATURE

Customer_ID identifies the row, but it does not explain customer behavior by itself.

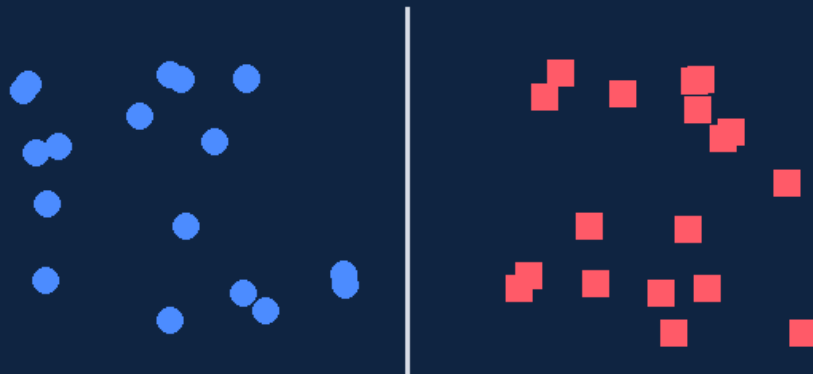
TWO COMMON TASKS

Classification vs. regression

Both are supervised learning; the output type is different.

CLASSIFICATION

Predict a category / class



Examples: churn/stay • fraud/not fraud •
approve/reject

REGRESSION

Predict a number



Examples: sales amount • wait time •
demand volume

TWO PHASES

Training vs. prediction

The model needs labels while learning, but not when scoring a new case.

TRAINING

Features + known label → model learns patterns

Example: historical customers + whether each churned

PREDICTION / INFERENCE

Features only → model outputs score / prediction

Example: current customer → churn probability 0.74

DATA USE

Training, validation, and test

A model should work on unseen examples — not just memorize the past.

Hold some data back so performance is measured honestly

TRAIN — learn the pattern

VALIDATE — tune
choices

TEST — final
unseen check

Think of it as study → practice exam → final exam. The test set should not guide training decisions.

Key idea: the test set is an honest final check. If it influences training decisions, reported performance can look better than reality.

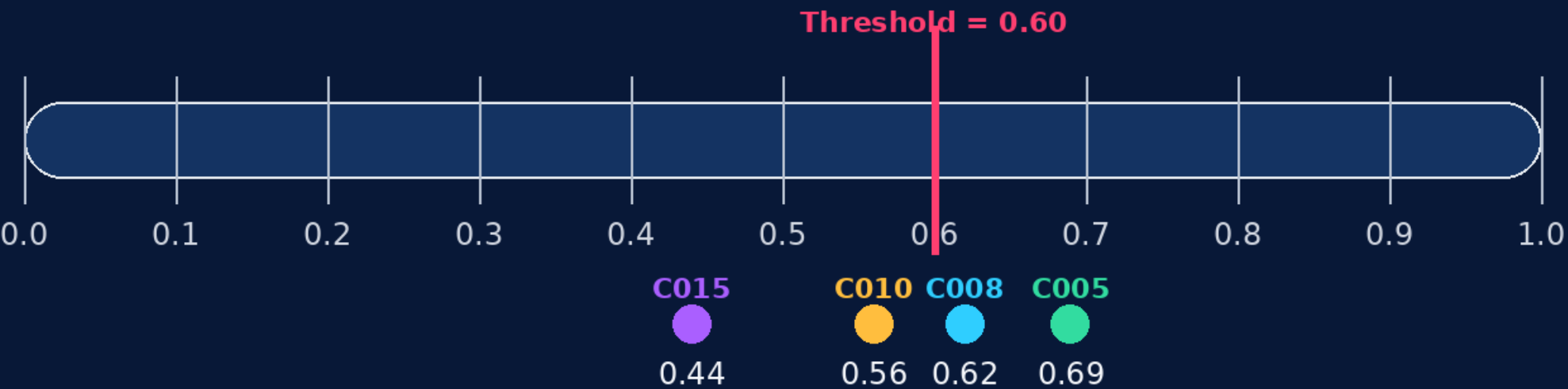
DECISION SUPPORT

Probability score + threshold

A score describes model confidence/risk; the threshold is a business policy for action.



Model score $\neq$ business action



EVALUATION

Why accuracy alone is not enough

False positives and false negatives create different business consequences.

Two models with the same accuracy can create very different business costs

		PREDICTED	
		Churn	Stay
ACTUAL	Churn	TRUE POSITIVE Correctly prioritize	FALSE NEGATIVE Miss a real churner
	Stay	FALSE POSITIVE Contact someone who would stay	TRUE NEGATIVE Correctly leave alone

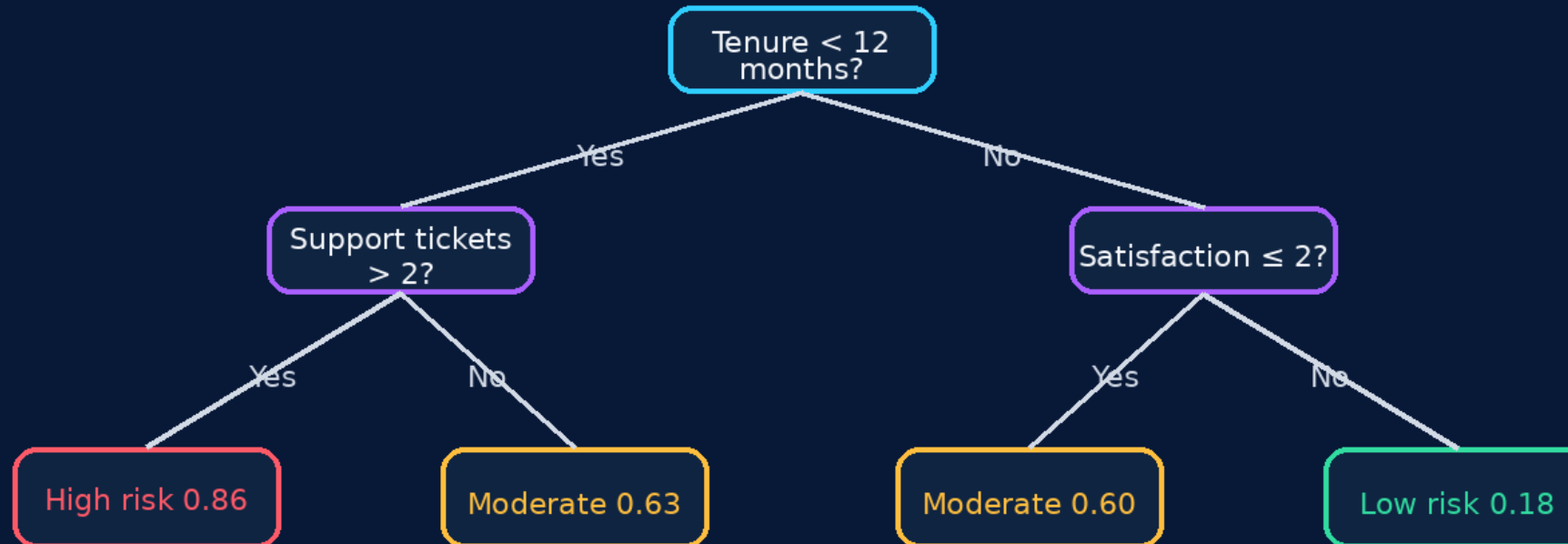
Business question: Which error costs more? • Missing a customer who will churn? • Spending retention effort on someone who would stay? The answer should influence threshold and workflow design.

MODEL INTUITION

Decision tree

A tree learns a sequence of if/then-style splits that separate cases.

Decision tree intuition: a chain of understandable splits

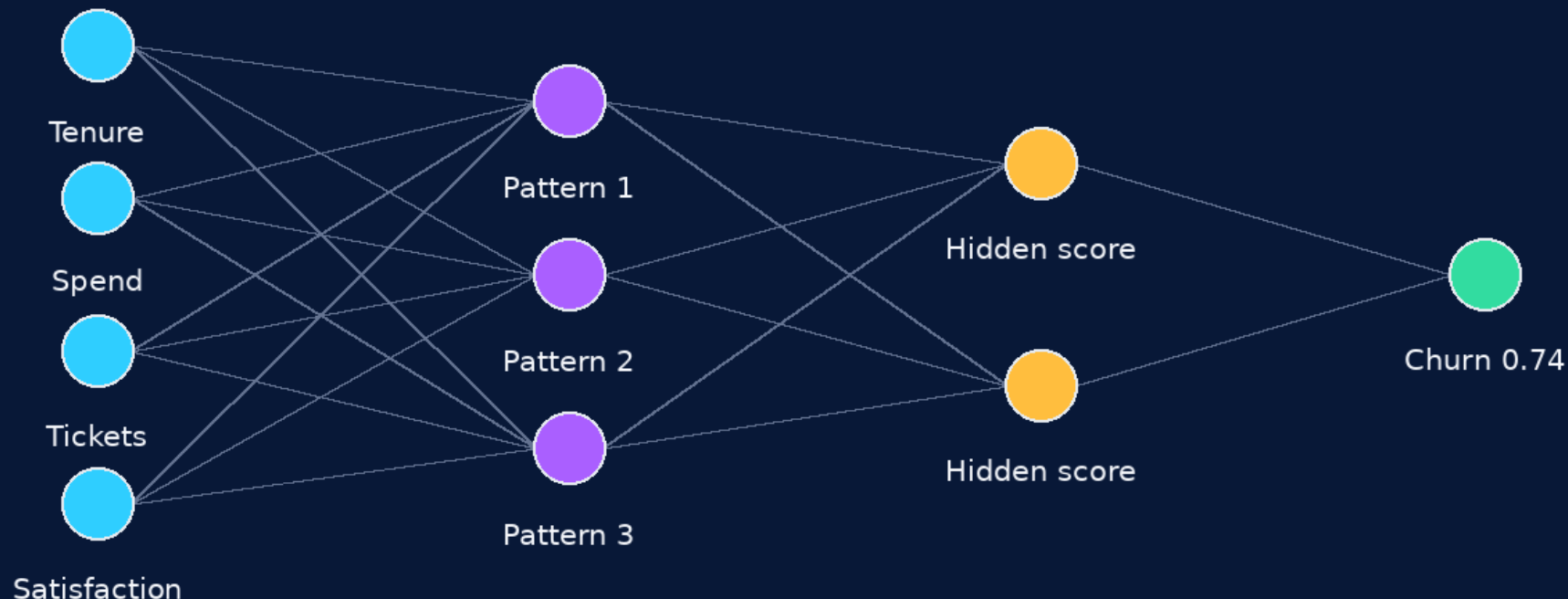


MODEL INTUITION

Neural network

Many weighted connections combine signals to learn more complex patterns.

Neural network intuition: many weighted connections combine signals



Today: understand the idea, not the math. Neural networks can learn complex patterns but are harder to explain than a simple tree.

BUSINESS USE CASES

Where supervised learning shows up

The same pattern — features → target → prediction — appears across business functions.

MARKETING

Churn • lead conversion •
campaign response

FINANCE / RISK

Fraud • default • credit risk

OPERATIONS

Delay • defect • demand
amount

HR

Attrition • hiring-support
outcomes

CUSTOMER SERVICE

Escalation • satisfaction risk

SUPPLY CHAIN

Stockout • supplier failure •
delivery time

QUICK CHECK

Can you identify the ML task?

Answer first; then explain your reasoning using output type, feature, and label.

A Invoice paid late?

B Days until shipment arrives?

C Monthly charge = ?

D Actual churn = ?

PART —

10-MINUTE BREAK

When we return: Applied Lab 1 — customer churn decision analysis

BE BACK IN 10

PART II

Applied Lab 1 — Customer Churn

Individual • no coding • interpret a pre-trained model and make a business recommendation.

BUSINESS QUESTION

Which customers should the retention team prioritize for outreach?

MODEL OUTPUT

Predicted churn probability for each current customer.

POLICY CONSTRAINT

Threshold = 0.60
Capacity = 8 contacts

LAB DATA

Read the table before touching the worksheet

The same row contains customer features, actual outcome, predicted score, and the class produced by the 0.60 policy.

Applied Lab 1 — sample from the customer churn dataset

ID	Tenure	Charge	Tickets	Satisfaction	Contract	Actual	Score	Flag?
C001	2	\$95	5	1	M-to-M	Yes	0.91	Yes
C004	3	\$69	2	3	M-to-M	No	0.72	Yes
C005	14	\$92	5	2	1-Year	Yes	0.69	Yes
C008	18	\$85	3	2	1-Year	Yes	0.62	Yes
C010	24	\$110	4	3	1-Year	Yes	0.56	No
C015	20	\$77	2	3	1-Year	Yes	0.44	No

LIVE DEMO

How to work through the lab

Follow this sequence; do not start by sorting randomly or guessing who “looks risky.”

- 1 Identify the six intended features + the label
- 2 Sort / inspect predicted churn probability
- 3 Apply the 0.60 threshold
- 4 Check capacity: exactly 8 flagged customers
- 5 Compare predictions with actual outcomes
- 6 Write a managerial recommendation with one control

demo focus: C005 (0.69) vs. C010 (0.56) — same model, different policy outcome.

LAB DEBRIEF

What did the 0.60 policy get wrong?

A good model can still make costly errors — and a useful threshold depends on the business context.

FALSE POSITIVES

C004 — score 0.72, actual = No

C007 — score 0.64, actual = No

Cost: retention effort may be spent on customers who would stay anyway.

FALSE NEGATIVES

C010 — score 0.56, actual = Yes

C015 — score 0.44, actual = Yes

Cost: a real churner may receive no proactive intervention.

MANAGERIAL LESSON

Use the model for prioritization, not automatic high-cost action. Monitor outcomes and revisit threshold as costs/capacity change.

The threshold is a policy choice. Its quality depends on model performance + business economics + operational capacity.

WEEK 4 SUBMISSIONS

Exactly what is graded this week

Keep these three items separate: lab, homework, and capstone milestone.

APPLIED LAB 1

Submit the completed individual worksheet in Canvas.

Focus: features/label • probabilities
• threshold • errors • managerial
recommendation.

CHAPTER 4 HOMEWORK

12 short-answer questions based
on the textbook + Week 4 lecture.

MILESTONE 1

Team repo: required files under /
docs + one-slide executive
summary.

Each of the 3 members individually
submits their own repo-access
screenshot + the shared repo
URL.

LEAVE WEEK 4 WITH FIVE IDEAS

Executive takeaway

Supervised learning is not “the model decides.” It is evidence → prediction → policy → action.

1 KNOWN OUTCOMES

Supervised learning learns from historical examples where the target is known.

2 FEATURES + LABEL

Features describe the case; the label is what we want to predict.

3 TASK TYPE

Classification predicts a category; regression predicts a number.

4 SCORE + THRESHOLD

A probability becomes an action only after the business defines a decision rule.

5 ERROR COST

False positives and false negatives matter differently depending on the business.

NEXT WEEK → Unsupervised Learning, Segmentation & Recommendations